Scientists Confirm an Ancient Asteroid Hit the North Sea

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About 43 to 46 million years ago, an **asteroid** crashed into the area now covered by the North Sea, sending up a **tsunami** more than 100 meters (330 feet) high. Now scientists say they finally have proof.

The impact left behind the Silverpit Crater, which lies about 700 meters below the seabed, roughly 80 miles off the coast of Yorkshire, England. The crater is about three kilometers wide and is surrounded by a 20-kilometer-wide ring of cracks in the rock.

Scientists discovered the crater in 2002. It has a round shape and a raised peak in the middle, like other **impact craters**. But some geologists thought it could have formed in other ways, such as from moving underground salt or a collapse caused by volcanoes. In 2009, geologists even held a vote on the question, and most rejected the **asteroid** idea.

A team led by Dr. Uisdean Nicholson of Heriot-Watt University in Edinburgh has now settled the debate. The researchers used new **seismic images**, which use sound waves to see layers of rock underground. They also studied rock pieces collected from a nearby oil well, and ran computer **simulations** of the crash.

The key clue was tiny crystals of "shocked" **quartz** and feldspar found at the same depth as the crater floor. These crystals have a pattern that can only be made by extreme pressure, like that of a space rock smashing into Earth.

"We were exceptionally lucky to find these," Nicholson said, calling it a "needle-in-a-haystack" effort. The study, published in Nature Communications, makes Silverpit one of the few confirmed **impact craters** on Earth.

Word Treasure

asteroid -- An asteroid is a rocky object that orbits the Sun and is smaller than a planet.

tsunami -- A tsunami is a series of huge ocean waves usually caused by a sudden push of water, such as an earthquake or impact.

seismic images -- Seismic images are pictures of underground rock layers made by sending sound waves into the ground and recording the echoes.

quartz -- Quartz is a common, hard mineral found in many kinds of rock.

simulations -- Simulations are computer models that recreate how something might have happened.

impact craters -- Impact craters are bowl-shaped holes made when an object from space smashes into a planet or moon.

Background you need

Earth has been hit by asteroids and comets throughout its history, but erosion, oceans and moving rock have erased or buried most of the craters.

Scientists confirm an impact crater by finding evidence that only an enormous collision can create, such as minerals changed by extreme shock pressure.

Seismic imaging lets geologists study buried structures, like Silverpit, by sending sound waves through rock and recording how they bounce back.

Before this study, some geologists thought salt movement or volcanic collapse could explain Silverpit's shape.

Break it down

LEAD: Scientists prove an asteroid hit the North Sea 43 to 46 million years ago, causing a huge tsunami

KEY FINDING: The Silverpit Crater was made by an impact

+- WHERE: about 700 meters below the seabed, roughly 80 miles off Yorkshire, England

+- SIZE: about three kilometers wide with a 20-kilometer ring of cracks

DEBATE: Found in 2002; some blamed salt or volcanoes; a 2009 vote rejected the asteroid idea

METHOD: Dr. Uisdean Nicholson's team used seismic images, oil well rock samples and computer simulations

EVIDENCE: Shocked quartz and feldspar crystals that only extreme pressure can make

RESULT: Silverpit joins the few confirmed impact craters on Earth, published in Nature Communications

Why it matters

Confirming impact craters helps scientists understand how often big space rocks have struck Earth and what effects, like giant tsunamis, they can cause.

Test what you remember

1. About how long ago did the asteroid hit?

- (A) 1 billion years ago
- (B) 4 to 6 thousand years ago
- (C) 66 million years ago
- (D) 43 to 46 million years ago

2. How tall was the tsunami?

- (A) Exactly 1 kilometer
- (B) About 10 meters
- (C) Less than 5 meters
- (D) More than 100 meters

3. When was the Silverpit Crater first discovered?

- (A) 2020
- (B) 2002
- (C) 2009
- (D) 1992

4. What did most geologists decide in a 2009 vote?

- (A) They said it was a volcano
- (B) They refused to vote
- (C) They rejected the asteroid idea
- (D) They supported the asteroid idea

5. What was the key clue that proved the impact?

- (A) Shocked quartz and feldspar crystals
- (B) Dinosaur bones
- (C) A piece of the asteroid
- (D) An old map

6. Where did the rock samples come from?

- (A) The Moon
- (B) A beach in Scotland
- (C) A nearby oil well
- (D) A museum

Answer key (try the quiz first!): 1: D 2: D 3: B 4: C 5: A 6: C

Questions to think about

1. Dr. Uisdean Nicholson's team: Says the shocked crystals prove the impact crater idea beyond doubt.

2. Earlier doubters: Some geologists argued underground salt movement or volcanic collapse formed the structure, and most rejected the asteroid idea in a 2009 vote.

MY THOUGHTS (at least 20 words):